

Big data analytic

Le Thanh Van

HCMC University of Technology

Dimensionality Reduction



Acknowledgement

2

- The lecture uses a portion of slides from “Mining of massive datasets” copyrighted by Jure Leskovec, Anand Rajaraman, Jeff Ullman, Stanford University.
- <http://www.mmds.org>

Curse of dimensionality

3

- **When data has too many features**

Dimensions grows, dimensions space increases exponentially

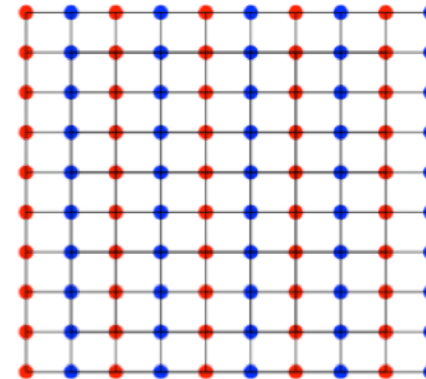
- **Disadvantages:**

- Sparsity of data
- Irrelevant data
- Complex and costly model for data analytic
- Difficulties in data clustering
- Difficult to visualize
- Difficult to find a solution
- Overfitting the model

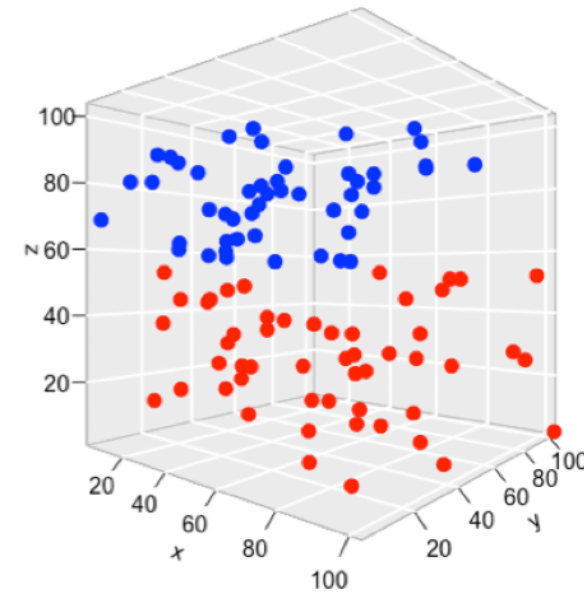
Source: Genomics



(A) 1-D



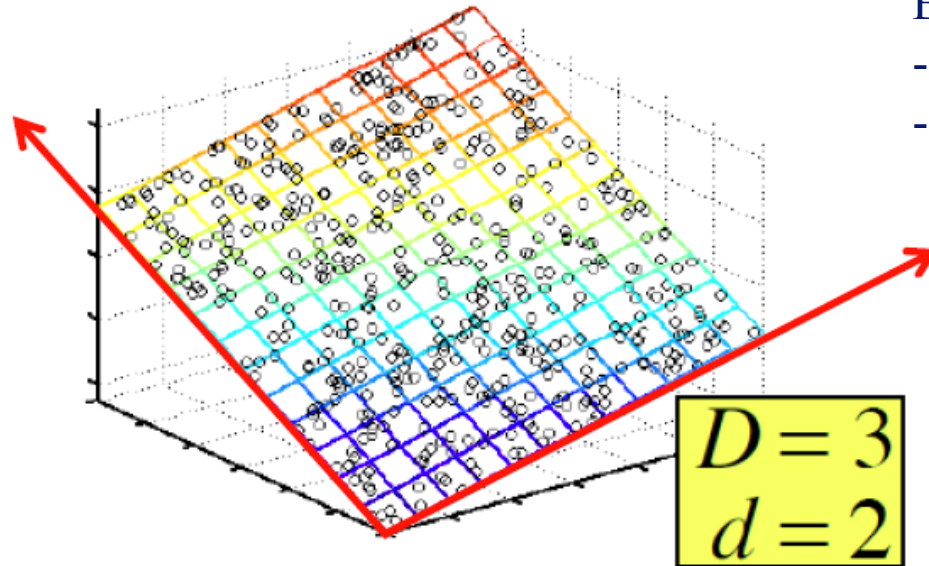
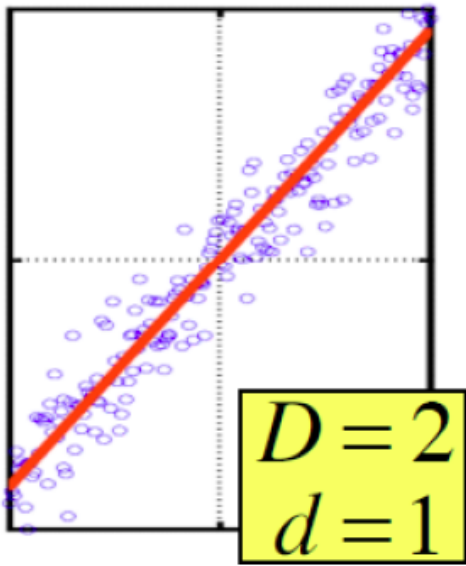
(B) 2-D



(C) 3-D

Dimensionality Reduction

4



Benefits:

- Better visualization detects more easily patterns
- Faster and more efficient model

- **Assumption:** Data lies on or near a low d -dimensional subspace
- **Axes of this subspace are effective representation of the data**

Dimensionality Reduction

5

- **Compress / reduce dimensionality:**
 - 10^6 rows; 10^3 columns; no updates
 - Random access to any cell(s); **small error: OK**

customer	day	We	Th	Fr	Sa	Su
		7/10/96	7/11/96	7/12/96	7/13/96	7/14/96
ABC Inc.		1	1	1	0	0
DEF Ltd.		2	2	2	0	0
GHI Inc.		1	1	1	0	0
KLM Co.		5	5	5	0	0
Smith		0	0	0	2	2
Johnson		0	0	0	3	3
Thompson		0	0	0	1	1

The above matrix is really “2-dimensional.” All rows can be reconstructed by scaling $[1 \ 1 \ 1 \ 0 \ 0]$ or $[0 \ 0 \ 0 \ 1 \ 1]$

Rank of a Matrix

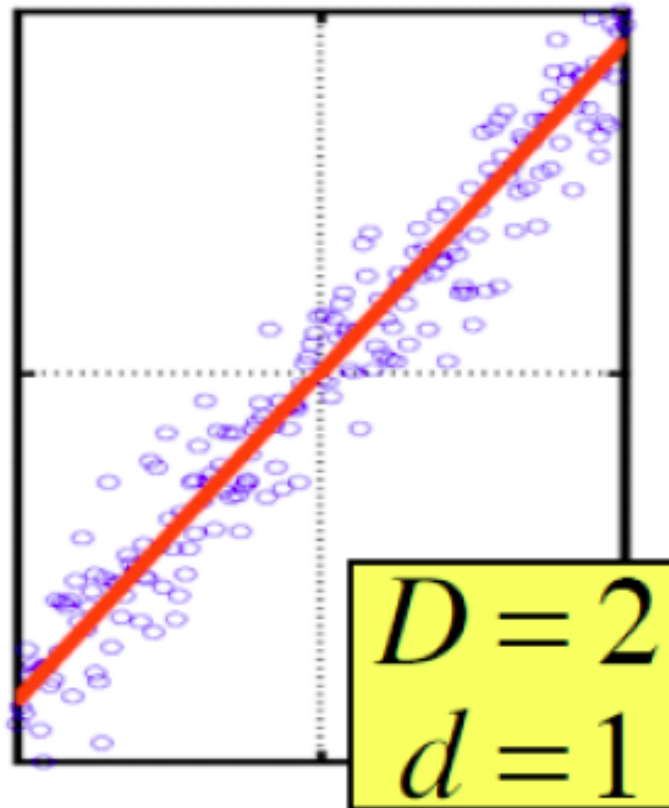
6

- **Q:** What is **rank** of a matrix **A**?
- **A:** Number of **linearly independent** columns/rows of **A**
- **For example:**
 - Matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix}$ has rank $r=2$
 - **Why?** The first two rows are linearly independent, so the rank is at least 2, but all three rows are linearly dependent (the first is equal to the sum of the second and third) so the rank must be less than 3.
- **Why do we care about low rank?**
 - We can write **A** as two “basis” vectors: $[1 \ 2 \ 1] \ [-2 \ -3 \ 1]$
 - And new coordinates of : $[1 \ 0] \ [0 \ 1] \ [1 \ 1]$

Dimensionality Reduction

7

- Goal of dimensionality reduction is to discover the axis of data!



Rather than representing every point with 2 coordinates we represent each point with 1 coordinate (corresponding to the position of the point on the red line).

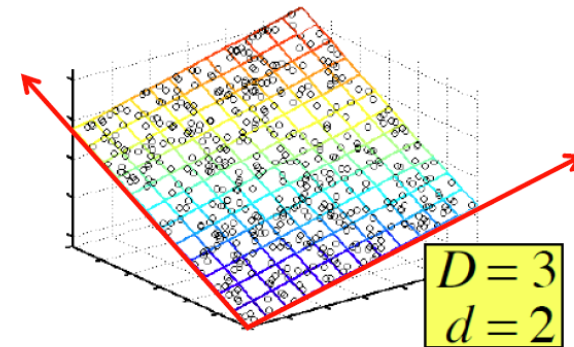
By doing this we incur a bit of **error** as the points do not exactly lie on the line

Why Reduce Dimensions?

8

Why reduce dimensions?

- **Discover hidden correlations/topics**
 - Words that occur commonly together
- **Remove redundant and noisy features**
 - Not all words are useful
- **Interpretation and visualization**
- **Easier storage and processing of the data**



PCA

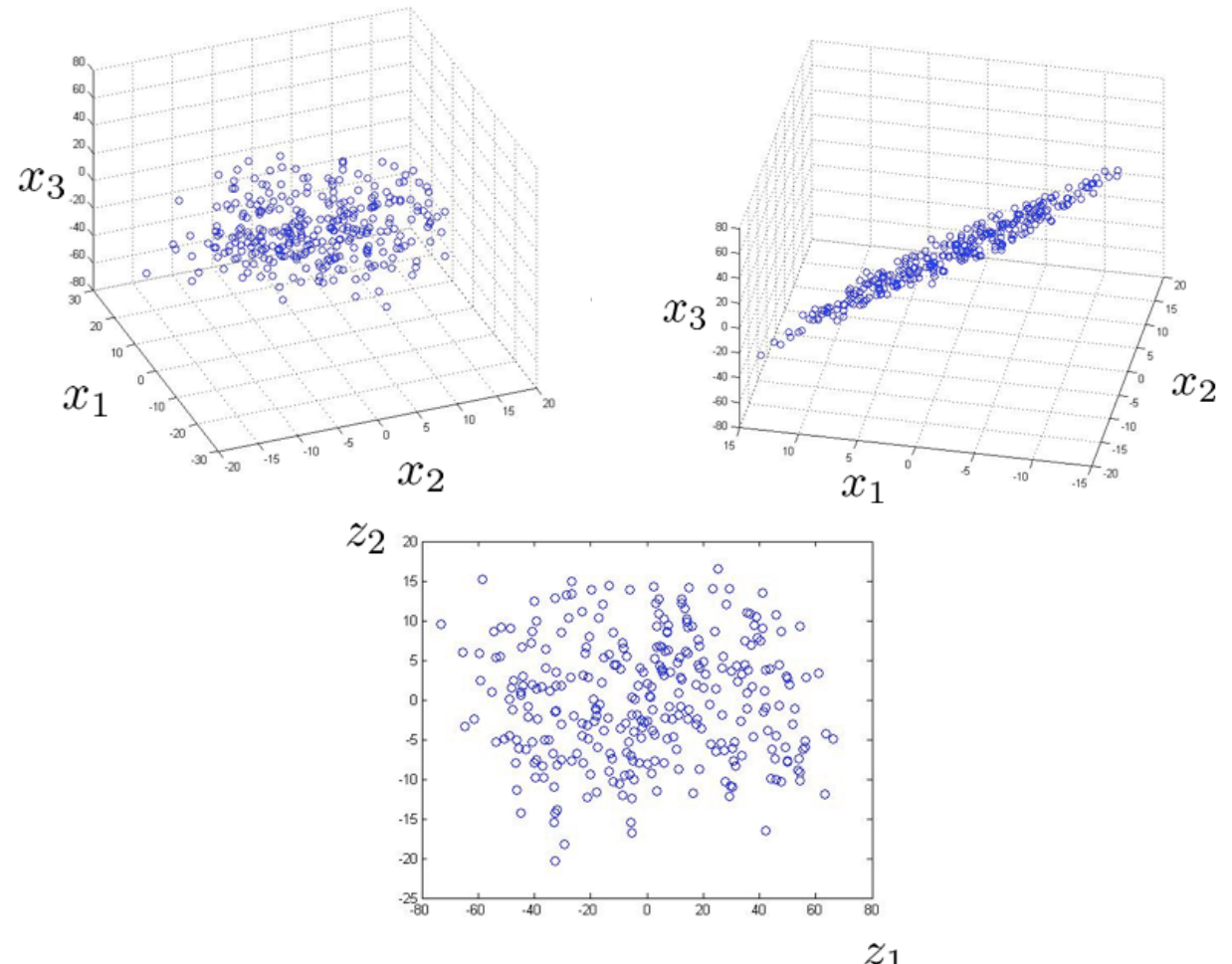
Principal Component Analysis



Principal Components Analysis (PCA)

10

- An exploratory technique used to reduce the dimensionality of the data set to lower dimension
- Can be used to:
 - Reduce number of dimensions in data
 - Find patterns in high-dimensional data
 - Visualize data of high dimensionality
- Example applications:
 - Face recognition
 - Image compression
 - Gene expression analysis



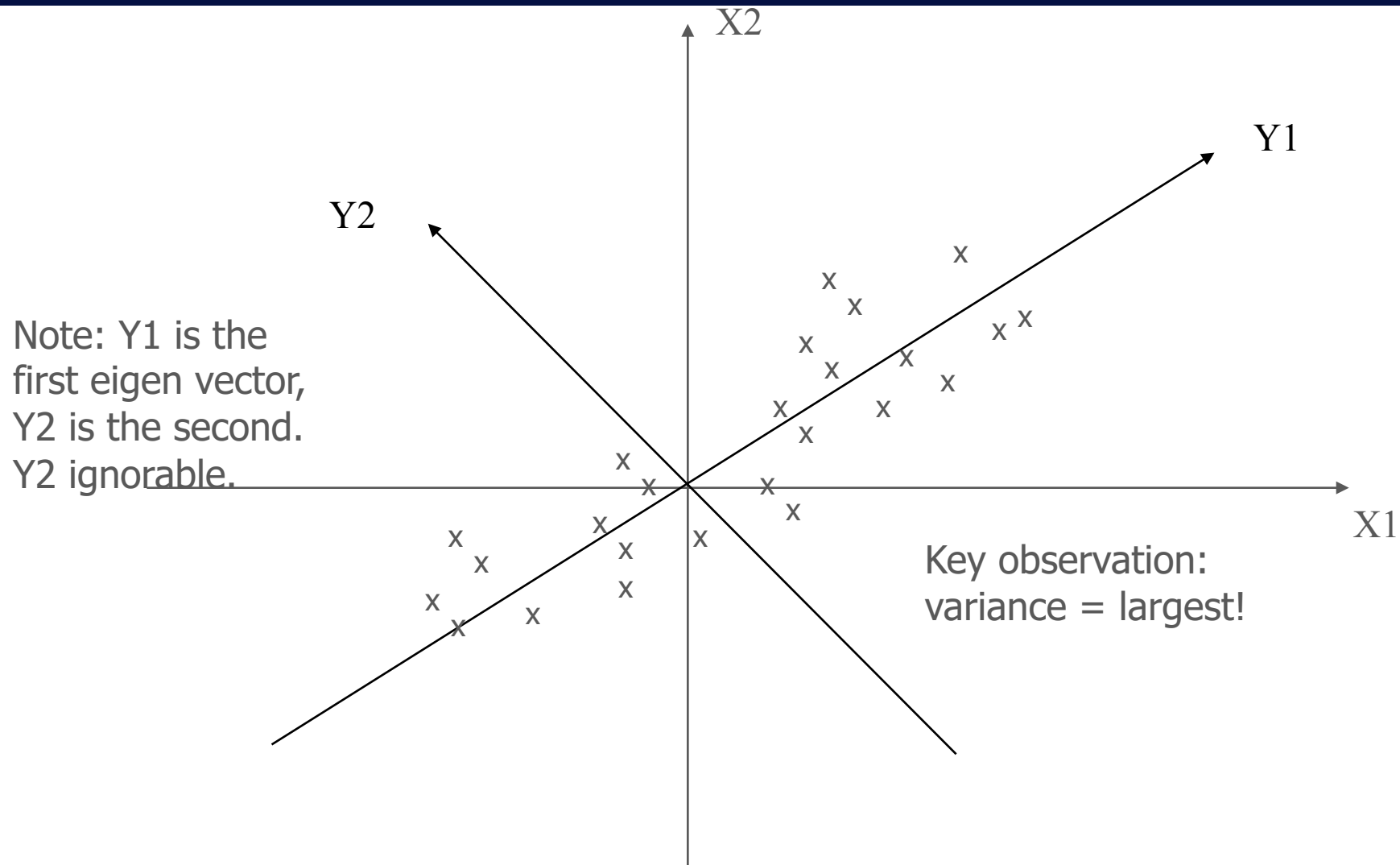
Principal Components Analysis Ideas

11

- Does the data set ‘span’ the whole of d dimensional space?
- For a matrix of m samples x n genes, create a new covariance matrix of size $n \times n$.
- Transform some large number of variables into a smaller number of uncorrelated variables called principal components (PCs).
- developed to capture as much of the variation in data as possible

Principal component analysis

12



Principal Component Analysis: one attribute first

13

- Question: how much spread is in the data along the axis? (distance to the mean)
- Variance=Standard deviation²

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

Temperature
42
40
24
30
15
18
15
30
15
30
35
30
40
20



Two dimensions

14

Covariance: measures the correlation between X and Y

- $\text{cov}(X,Y)=0$: independent
- $\text{Cov}(X,Y)>0$: move same direction
- $\text{Cov}(X,Y)<0$: move opposite direction

$$\text{cov}(X,Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{(n - 1)}$$

How about 3 variables?

X=Temperature	Y=Humidity
40	90
40	90
40	90
30	90
15	70
15	70
15	70
30	90
15	70
30	70
30	70
30	70
30	90
40	70
30	90

PCA - Definition

15

- M : square matrix, λ : a constant, e : non-zero column vector of rows as M .
- λ : eigenvalue of M , Vectors $\mathbf{x}$ having same direction as $A\mathbf{x}$ are called *eigenvectors* of A
- e : eigenvector of M if $Me = \lambda e$ (e having same direction as Me)
- Eigen vector must be unit vector, $\sum e_{ij}^2 = 1$, $e_{11} > 0$.

$$\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} = 7 \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$\uparrow$
M

$\uparrow$ $\uparrow$
Eigenvalue **Eigenvector**

Eigenvalues & eigenvectors

16

- $\mathbf{M}\mathbf{e} = \lambda\mathbf{e} \Leftrightarrow (\mathbf{M} - \lambda\mathbf{I})\mathbf{e} = \mathbf{0}$
- How to calculate $\mathbf{e}$ and λ :
 - Calculate $\det(\mathbf{M} - \lambda\mathbf{I})$, yields a polynomial (degree n)
 - Determine roots to $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$, roots are eigenvalues λ
 - Solve $(\mathbf{M} - \lambda\mathbf{I})\mathbf{e} = \mathbf{0}$ for each λ to obtain eigenvectors $\mathbf{e}$

Principal components

17

1. principal component (PC1)

The eigenvalue with the largest absolute value will indicate that the data have the largest variance along its eigenvector (most "spread out")

- the direction along which there is greatest variation

2. principal component (PC2)

The direction with maximum variation left in data, orthogonal to the PC1.

- In general, only few directions manage to capture most of the variability in the data.

PCA: try to Project high-dimensional data onto the most important axes

Steps of PCA

- For matrix M , vectors $\mathbf{e}$ (=column vector) having same direction as $M\mathbf{e}$:
 - *eigenvectors* of M is $\mathbf{e}$ such that $M\mathbf{e}=\lambda\mathbf{e}$,
 - λ is called an *eigenvalue* of C .
- $M\mathbf{e}=\lambda\mathbf{e} \Leftrightarrow (M-\lambda I)\mathbf{e}=0$
- Calculate $\det(M-\lambda I)$
- Find λ such that $\det(M-\lambda I) = 0$
- Replace λ in equations from $M\mathbf{e}=\lambda\mathbf{e}$ to find $\mathbf{e}$
- Divide each component of $\mathbf{e}$ by **sum of squares of all components** to derive the unit vector

PCA example

19

$$M = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \quad M - \lambda I = \begin{bmatrix} 3 - \lambda & 2 \\ 2 & 6 - \lambda \end{bmatrix}$$

$$\det(M - \lambda I) = \lambda^2 - 9\lambda + 14$$
$$\det(.) = 0, \lambda = 7, \lambda = 2$$

$$e = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 7 \begin{bmatrix} x \\ y \end{bmatrix}$$

$$Me = \lambda e$$

$$3x + 2y = 7x$$

$$2x + 6y = 7y$$

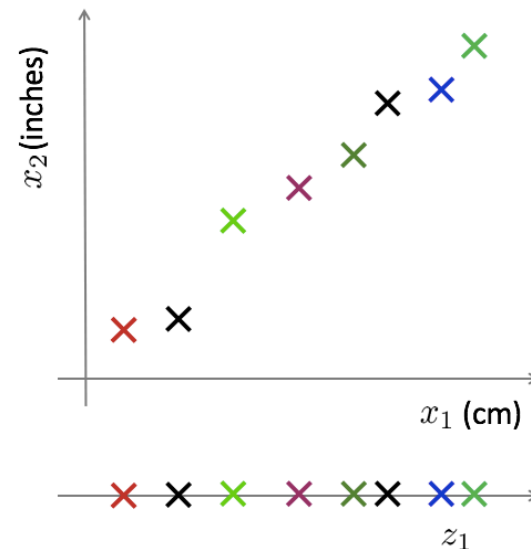
$$y = 2x$$

$$\text{Possible } e = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{Unit vector } e (\lambda = 7) = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \quad e (\lambda = 2) = \begin{bmatrix} 2/\sqrt{5} \\ -1/\sqrt{5} \end{bmatrix}$$

PCA in dimensionality reduction

20

- Goals: Find k -dimension projection that best preserves variance
- Compute eigenvalues λ and eigenvectors $\mathbf{e}$ of $\mathbf{M}^T\mathbf{M}$
 - $\mathbf{E}$: matrix of eigenvectors - rigid rotation in a high dimensional space
- Select k first columns of $\mathbf{E}$ and project points onto subspace spanned $\mathbf{ME}_k$



Example

21

$$\mathbf{M} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 4 \\ 4 & 3 \end{bmatrix} \quad \mathbf{A} = \mathbf{M}^T \mathbf{M} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 4 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 30 & 28 \\ 28 & 30 \end{bmatrix}$$
$$(30 - \lambda)(30 - \lambda) - 28 \times 28 = 0 \quad \lambda = 58, \lambda = 2$$

$$\begin{bmatrix} 30 & 28 \\ 28 & 30 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 58 \begin{bmatrix} x \\ y \end{bmatrix} \quad \mathbf{e}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} 30 & 28 \\ 28 & 30 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2 \begin{bmatrix} x \\ y \end{bmatrix} \quad \mathbf{e}_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad \mathbf{ME}_1 = \begin{bmatrix} 3/\sqrt{2} \\ 3/\sqrt{2} \\ 7/\sqrt{2} \\ 7/\sqrt{2} \end{bmatrix}$$

$$\mathbf{A}\mathbf{e} = \lambda\mathbf{e}$$

Power Iteration

22

- Choose a random vector $\mathbf{x}_0$

- Calculate

$$\mathbf{x}_{k+1} := \frac{M\mathbf{x}_k}{\|M\mathbf{x}_k\|}$$

Frobenius norm:

$$\|M\|_F = \sqrt{\sum_{ij} M_{ij}^2}$$

- $\|N\|$: Frobenius norm – square root of the sum of square of the elements of N
- Multiply $\mathbf{x}_k$ until convergence $\|\mathbf{x}_k - \mathbf{x}_{k+1}\| < \text{threshold}$
- $\mathbf{x} = \text{final } \mathbf{x}_k$, $\mathbf{x}$ is approximately the principal eigenvector of M
- Compute $\lambda_1 = \mathbf{x}^T M \mathbf{x}$ ($M\mathbf{x} = \lambda \mathbf{x}$)

For second eigenpair, create $M^* = M - \lambda_1 \mathbf{x} \mathbf{x}^T$

- Apply Power Iteration on M^* to compute the second largest eigenvalue

SVD

Singular vector decomposition



SVD – Definition (Singular Value Decomposition)

24

$$\mathbf{A}_{[m \times n]} = \mathbf{U}_{[m \times r]} \mathbf{\Sigma}_{[r \times r]} (\mathbf{V}_{[n \times r]})^T$$

- **A: Input data matrix**
 - $m \times n$ matrix (e.g., m documents, n terms)
- **U: Left singular vectors**
 - $m \times r$ matrix (m documents, r concepts)
- **Σ : Singular values**
 - $r \times r$ diagonal matrix (strength of each ‘concept’)
(r : rank of the matrix **A**)
- **V: Right singular vectors**
 - $n \times r$ matrix (n terms, r concepts)

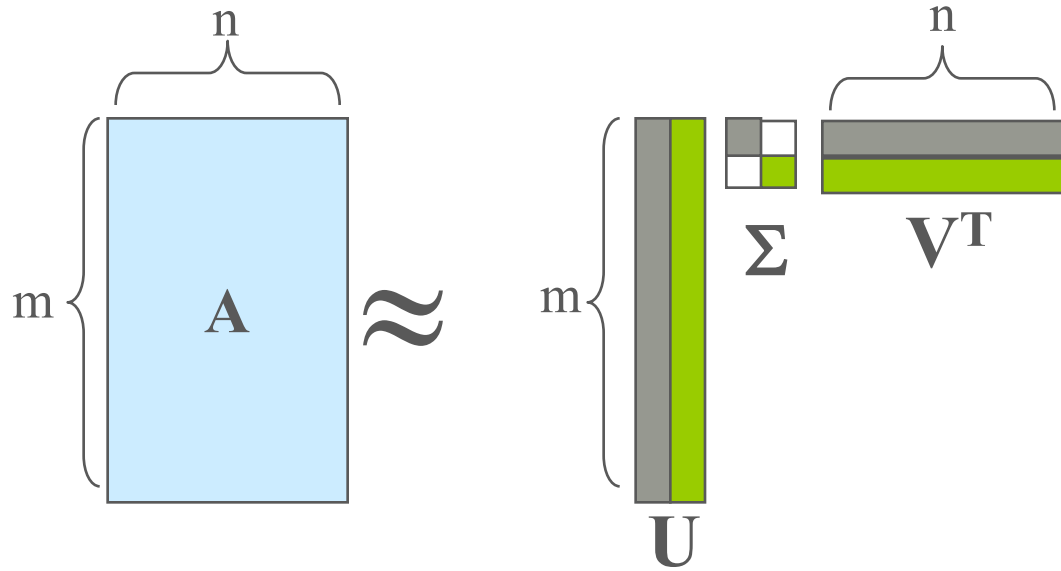
Produce a low-dimensional representation of a high-dimensional matrix

- eliminate the less important parts,
- produce an approximate representation with any desired number of dimensions

SVD

25

$$\mathbf{A} \approx \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \sum_i \sigma_i \mathbf{u}_i \circ \mathbf{v}_i^T$$



$\mathbf{A} =$

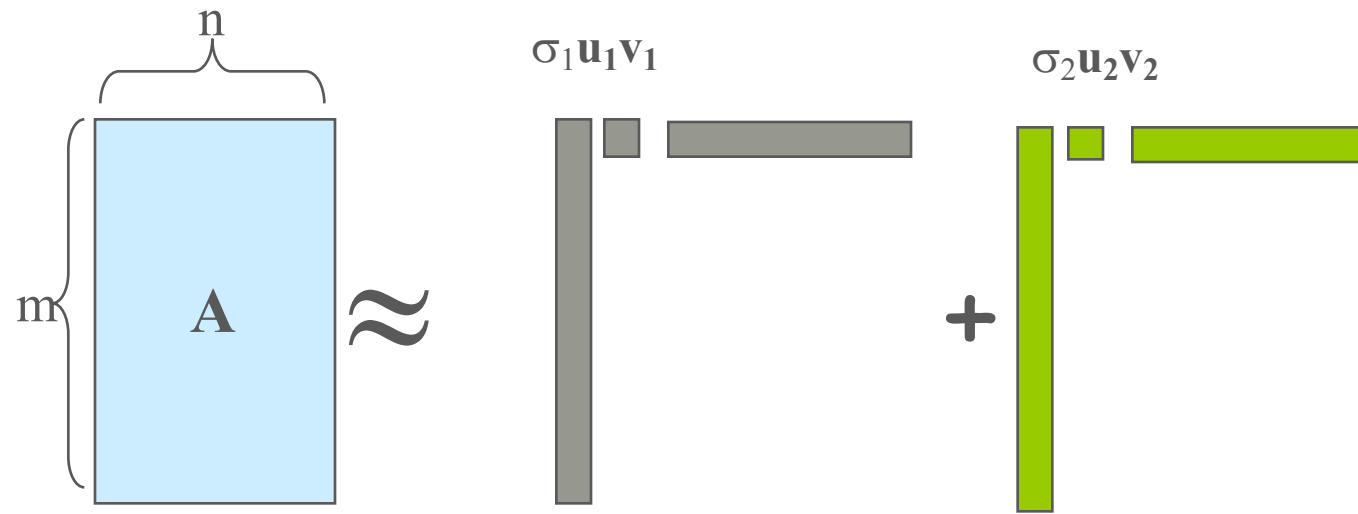
	Matrix	Alien	Star Wars	Casablanca	Titanic
Joe	1	1	1	0	0
Jim	3	3	3	0	0
John	4	4	4	0	0
Jack	5	5	5	0	0
Jill	0	0	0	4	4
Jenny	0	0	0	5	5
Jane	0	0	0	2	2

$$\approx \begin{bmatrix} .14 & 0 \\ .42 & 0 \\ .56 & 0 \\ .70 & 0 \\ 0 & .60 \\ 0 & .75 \\ 0 & .30 \end{bmatrix} \begin{bmatrix} 12.4 & 0 \\ 0 & 9.5 \end{bmatrix} \begin{bmatrix} .58 & .58 & .58 & 0 & 0 \\ 0 & 0 & 0 & .71 & .71 \end{bmatrix}$$

SVD

26

$$\mathbf{A} \approx \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \sum_i \sigma_i \mathbf{u}_i \circ \mathbf{v}_i^T$$



σ_i ... scalar

$\mathbf{u}_i$... vector

$\mathbf{v}_i$... vector

SVD - Properties

27

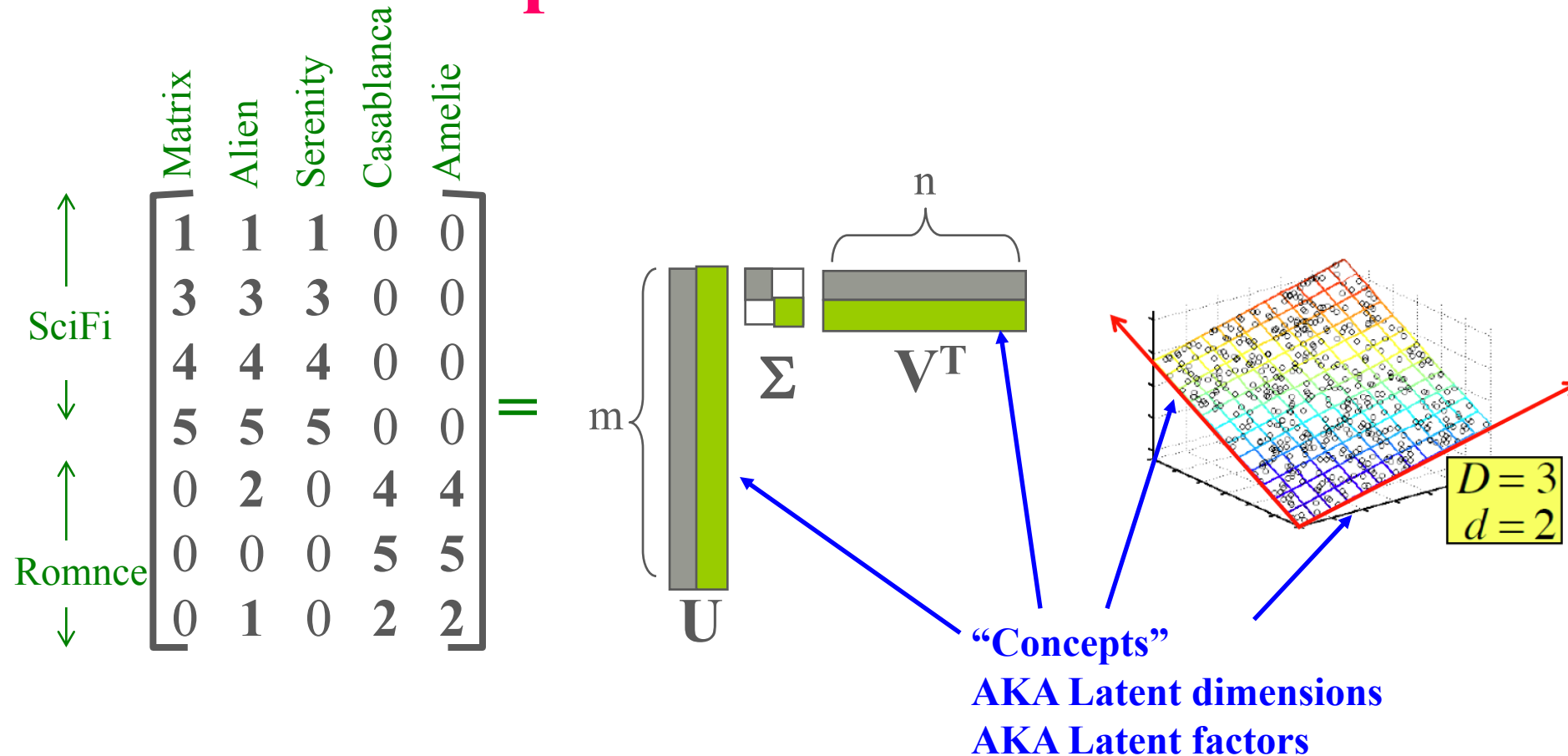
It is **always** possible to decompose a real matrix A into $A = U \Sigma V^T$, where

- U, Σ, V : **unique**
- U, V : **column orthonormal**
 - $U^T U = I; V^T V = I$ (I : identity matrix)
 - (Columns are orthogonal unit vectors)
- Σ : **diagonal**
 - Entries (**singular values**) are **positive**, and sorted in decreasing order ($\sigma_1 \geq \sigma_2 \geq \dots \geq 0$): express the important weights

SVD – Example: Users-to-Movies

28

- $A = U \Sigma V^T$ - example: Users to Movies



SVD – Example: Users-to-Movies

29

- $A = U \Sigma V^T$ - example: Users to Movies

$$\begin{array}{c}
 \begin{array}{c} \uparrow \\ \text{SciFi} \\ \downarrow \\ \uparrow \\ \text{Romnce} \\ \downarrow \end{array}
 \end{array}
 \begin{array}{c}
 \text{Matrix} \\
 \text{Alien} \\
 \text{Serenity} \\
 \text{Casablanca} \\
 \text{Amelie}
 \end{array}
 \begin{bmatrix}
 1 & 1 & 1 & 0 & 0 \\
 3 & 3 & 3 & 0 & 0 \\
 4 & 4 & 4 & 0 & 0 \\
 5 & 5 & 5 & 0 & 0 \\
 0 & 2 & 0 & 4 & 4 \\
 0 & 0 & 0 & 5 & 5 \\
 0 & 1 & 0 & 2 & 2
 \end{bmatrix}
 =
 \begin{bmatrix}
 0.13 & 0.02 & -0.01 \\
 0.41 & 0.07 & -0.03 \\
 0.55 & 0.09 & -0.04 \\
 0.68 & 0.11 & -0.05 \\
 0.15 & -0.59 & 0.65 \\
 0.07 & -0.73 & -0.67 \\
 0.07 & -0.29 & 0.32
 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix}
 12.4 & 0 & 0 \\
 0 & 9.5 & 0 \\
 0 & 0 & 1.3
 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix}
 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
 0.40 & -0.80 & 0.40 & 0.09 & 0.09
 \end{bmatrix}$$

SVD – Example: Users-to-Movies

30

• $A = U \Sigma V^T$ - example: Users to Movies

Diagram illustrating the SVD decomposition of a user-movie rating matrix A into three matrices: U , Σ , and V^T .

Matrix A (Users to Movies):

	Matrix	Alien	Serenity	Casablanca	Amelie
SciFi	1	1	1	0	0
	3	3	3	0	0
	4	4	4	0	0
	5	5	5	0	0
Romance	0	2	0	4	4
	0	0	0	5	5
	0	1	0	2	2

Matrix Σ (Concepts):

	SciFi-concept	Romance-concept	
	0.13	0.02	-0.01
	0.41	0.07	-0.03
	0.55	0.09	-0.04
	0.68	0.11	-0.05
	0.15	-0.59	0.65
	0.07	-0.73	-0.67
	0.07	-0.29	0.32

Matrix V^T (Concepts):

12.4	0	0
0	9.5	0
0	0	1.3

Matrix U (Users):

0.56	0.59	0.56	0.09	0.09
0.12	-0.02	0.12	-0.69	-0.69
0.40	-0.80	0.40	0.09	0.09

The decomposition is shown as:

$$A = U \Sigma V^T$$

SVD – Example: Users-to-Movies

31

- $A = U \Sigma V^T$ - example:

U is “user-to-concept”
similarity matrix

$$\begin{array}{c}
 \begin{array}{c} \uparrow \\ \text{SciFi} \\ \downarrow \\ \uparrow \\ \text{Romnce} \\ \downarrow \end{array}
 \begin{bmatrix}
 \text{Matrix} & \text{Alien} & \text{Serenity} & \text{Casablanca} & \text{Amelie} \\
 1 & 1 & 1 & 0 & 0 \\
 3 & 3 & 3 & 0 & 0 \\
 4 & 4 & 4 & 0 & 0 \\
 5 & 5 & 5 & 0 & 0 \\
 0 & 2 & 0 & 4 & 4 \\
 0 & 0 & 0 & 5 & 5 \\
 0 & 1 & 0 & 2 & 2
 \end{bmatrix}
 =
 \begin{array}{c}
 \text{SciFi-concept} \quad \text{Romance-concept} \\
 \downarrow \quad \swarrow \\
 \begin{bmatrix}
 0.13 & 0.02 & -0.01 \\
 0.41 & 0.07 & -0.03 \\
 0.55 & 0.09 & -0.04 \\
 0.68 & 0.11 & -0.05 \\
 0.15 & -0.59 & 0.65 \\
 0.07 & -0.73 & -0.67 \\
 0.07 & -0.29 & 0.32
 \end{bmatrix}
 \end{array}
 \mathbf{X}
 \begin{bmatrix}
 12.4 & 0 & 0 \\
 0 & 9.5 & 0 \\
 0 & 0 & 1.3
 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix}
 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
 0.40 & -0.80 & 0.40 & 0.09 & 0.09
 \end{bmatrix}
 \end{array}$$

SVD – Example: Users-to-Movies

32

• $A = U \Sigma V^T$ - example:

Diagram illustrating the SVD decomposition of a user-movie rating matrix A into three matrices: U , Σ , and V^T .

Matrix A (User-Movie Ratings):

	Matrix	Alien	Serenity	Casablanca	Amelie
SciFi	1	1	1	0	0
	3	3	3	0	0
	4	4	4	0	0
	5	5	5	0	0
	0	2	0	4	4
Romnce	0	0	0	5	5
	0	1	0	2	2

Matrix Σ (Concepts):

	SciFi-concept		
	0.13	0.02	-0.01
	0.41	0.07	-0.03
	0.55	0.09	-0.04
	0.68	0.11	-0.05
	0.15	-0.59	0.65
	0.07	-0.73	-0.67
	0.07	-0.29	0.32

Matrix V^T (Concept Strengths):

	SciFi	Romnce	
	12.4	0	0
	0	9.5	0
	0	0	1.3

Matrix U (User Latent Factors):

0.56	0.59	0.56	0.09	0.09
0.12	-0.02	0.12	-0.69	-0.69
0.40	-0.80	0.40	0.09	0.09

The decomposition is shown as:

$$A = U \Sigma V^T$$

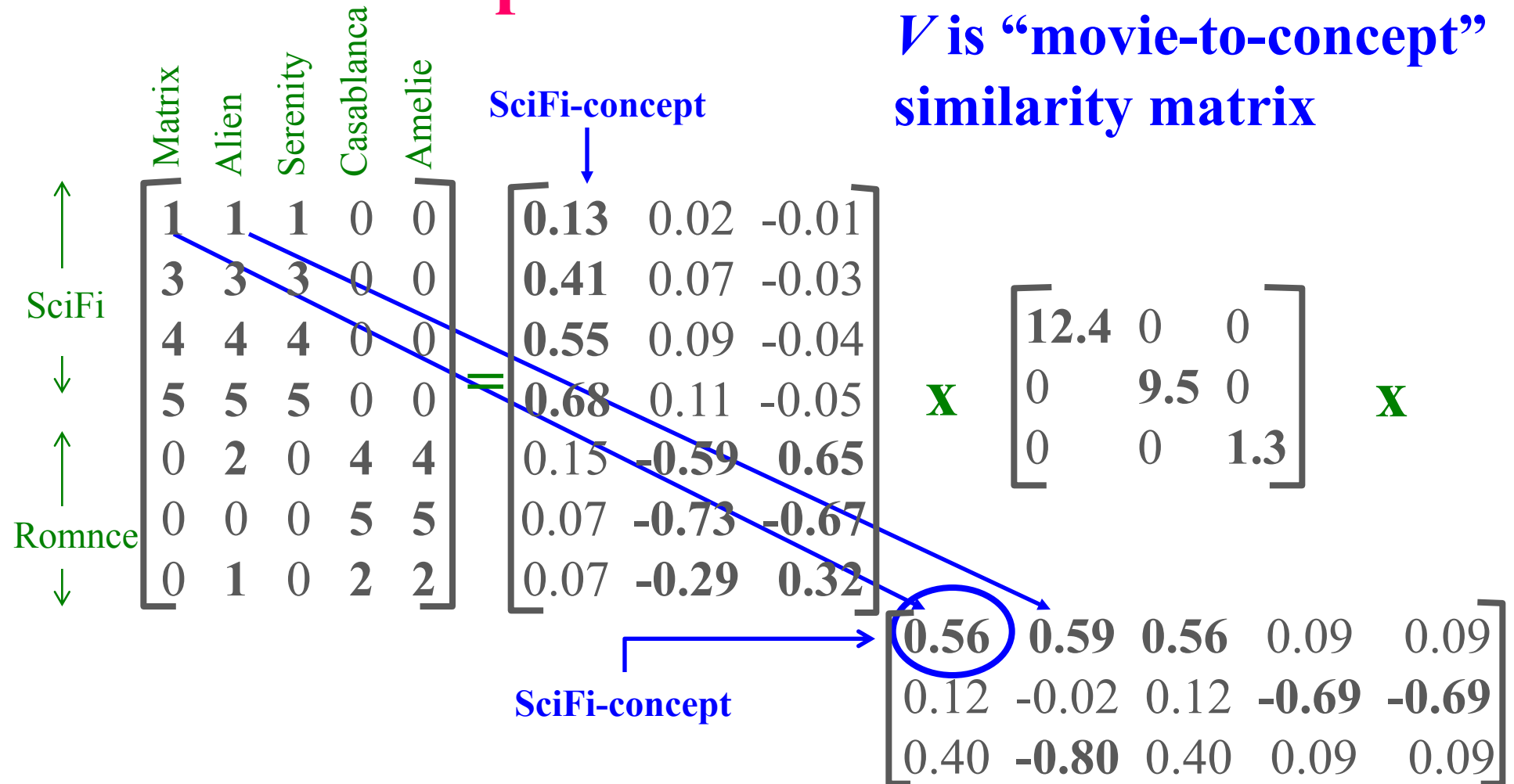
Annotations:

- Green arrows on the left of matrix A indicate the "SciFi" and "Romnce" concepts.
- A blue arrow points to the "SciFi-concept" column in matrix Σ .
- A blue circle highlights the value 12.4 in matrix V^T , with a label "strength" of the SciFi-concept.

SVD – Example: Users-to-Movies

33

- $A = U \Sigma V^T$ - example:



SVD - Interpretation #1

34

‘**movies**’, ‘**users**’ and ‘**concepts**’:

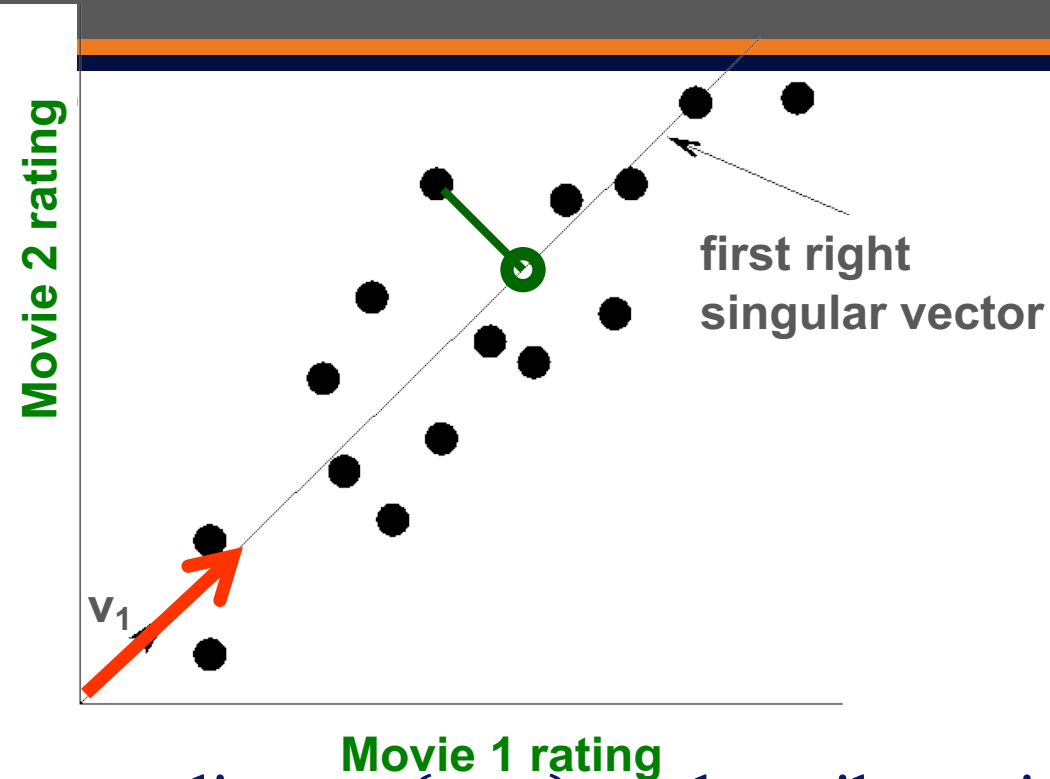
- U : user-to-concept similarity matrix
- V : movie-to-concept similarity matrix
- Σ : its diagonal elements: ‘strength’ of each concept

Dimensionality Reduction with SVD



SVD – Dimensionality Reduction

36



- Instead of using two coordinates (x, y) to describe point locations, let's use only one coordinate (z)
- Point's position is its location along vector v_1
- How to choose v_1 ? **Minimize reconstruction error**

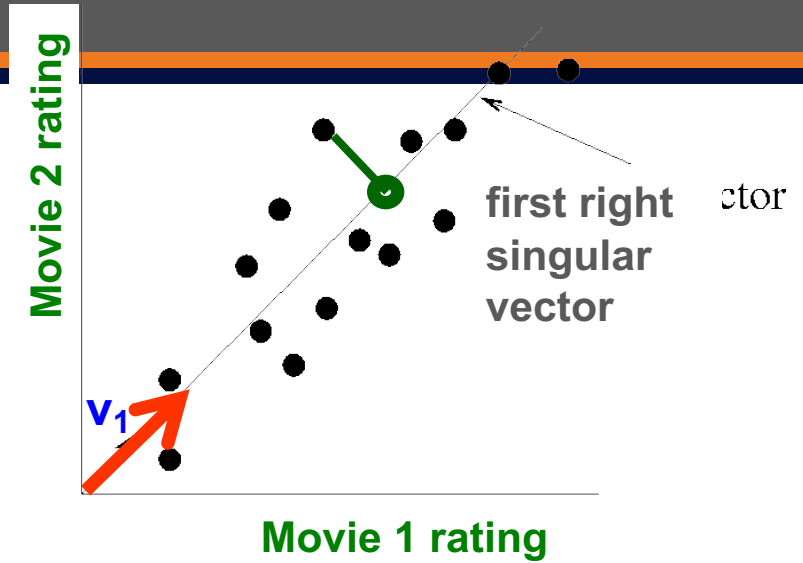
SVD – Dimensionality Reduction

37

- **Goal:** Minimize the sum of reconstruction errors:

$$\sum_{i=1}^N \sum_{j=1}^D \|x_{ij} - z_{ij}\|^2$$

- where x_{ij} are the “old” and z_{ij} are the “new” coordinates
- **SVD gives ‘best’ axis to project on:**
 - ‘best’ = minimizing the reconstruction errors
- In other words, **minimum reconstruction error**



SVD - Interpretation #2

38

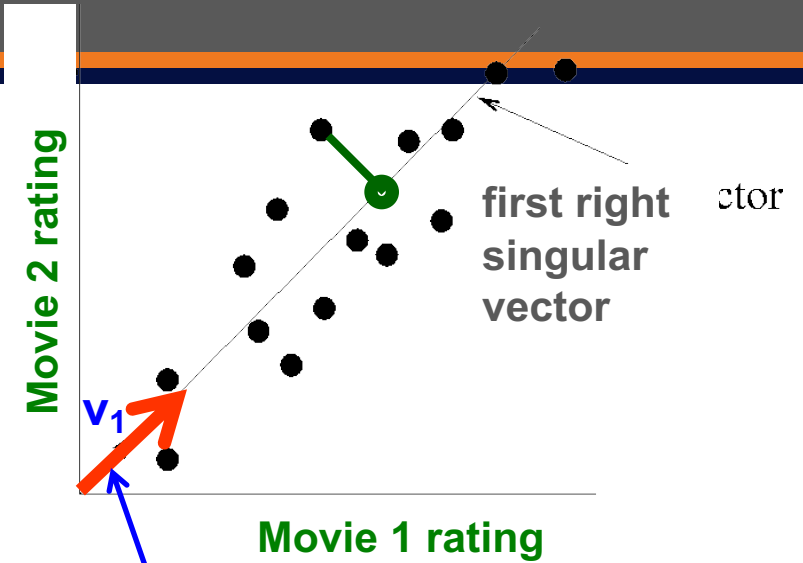
• $A = U \Sigma V^T$ - example:

- V : “movie-to-concept” matrix
- U : “user-to-concept” matrix

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{x}$$

$$\begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \mathbf{x}$$

$$\begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$



SVD - Interpretation #2

39

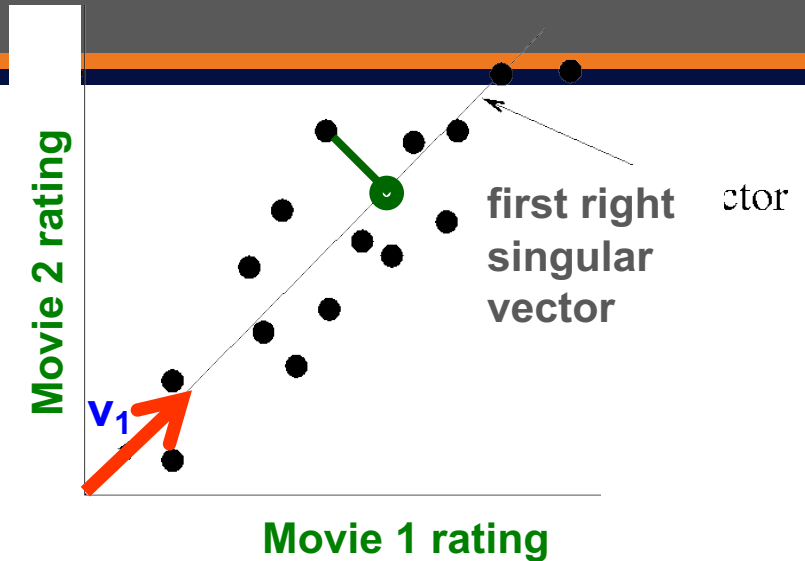
- $A = U \Sigma V^T$ - example:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X}$$

$$\begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \mathbf{X}$$

$$\begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

variance ('spread')
on the v_1 axis



SVD - Interpretation #2

40

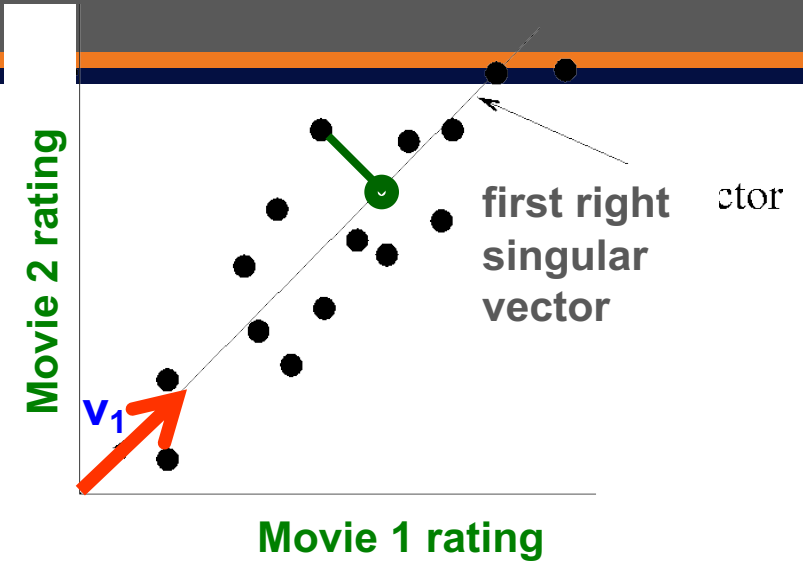
$A = U \Sigma V^T$ - example:

- $U \Sigma$: Gives the coordinates of the points in the projection axis

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix}$$

Projection of users
on the “Sci-Fi” axis
($U \Sigma$)^T:

$$\begin{bmatrix} 1.61 & 0.19 & -0.01 \\ 5.08 & 0.66 & -0.03 \\ 6.82 & 0.85 & -0.05 \\ 8.43 & 1.04 & -0.06 \\ 1.86 & -5.60 & 0.84 \\ 0.86 & -6.93 & -0.87 \\ 0.86 & -2.75 & 0.41 \end{bmatrix}$$



SVD - Interpretation #2

41

More details

- **Q:** How exactly is dim. reduction done?

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X} \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \mathbf{X} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

SVD - Interpretation #2

42

More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X} \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & \cancel{1.3} \end{bmatrix} \mathbf{X} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

SVD - Interpretation #2

43

More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X} \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & \cancel{1.3} \end{bmatrix} \mathbf{X} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

SVD - Interpretation #2

More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \mathbf{X} \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \mathbf{X} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

SVD - Interpretation #2

45

More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.13 & 0.02 \\ 0.41 & 0.07 \\ 0.55 & 0.09 \\ 0.68 & 0.11 \\ 0.15 & -0.59 \\ 0.07 & -0.73 \\ 0.07 & -0.29 \end{bmatrix} \mathbf{X} \begin{bmatrix} 12.4 & 0 \\ 0 & 9.5 \end{bmatrix} \mathbf{X} \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \end{bmatrix}$$

SVD - Interpretation #2

46

More details

- **Q:** How exactly is dim. reduction done?
- **A:** Set smallest singular values to zero

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.92 & 0.95 & 0.92 & 0.01 & 0.01 \\ 2.91 & 3.01 & 2.91 & -0.01 & -0.01 \\ 3.90 & 4.04 & 3.90 & 0.01 & 0.01 \\ 4.82 & 5.00 & 4.82 & 0.03 & 0.03 \\ 0.70 & 0.53 & 0.70 & 4.11 & 4.11 \\ -0.69 & 1.34 & -0.69 & 4.78 & 4.78 \\ 0.32 & 0.23 & 0.32 & 2.01 & 2.01 \end{bmatrix}$$

Frobenius norm:

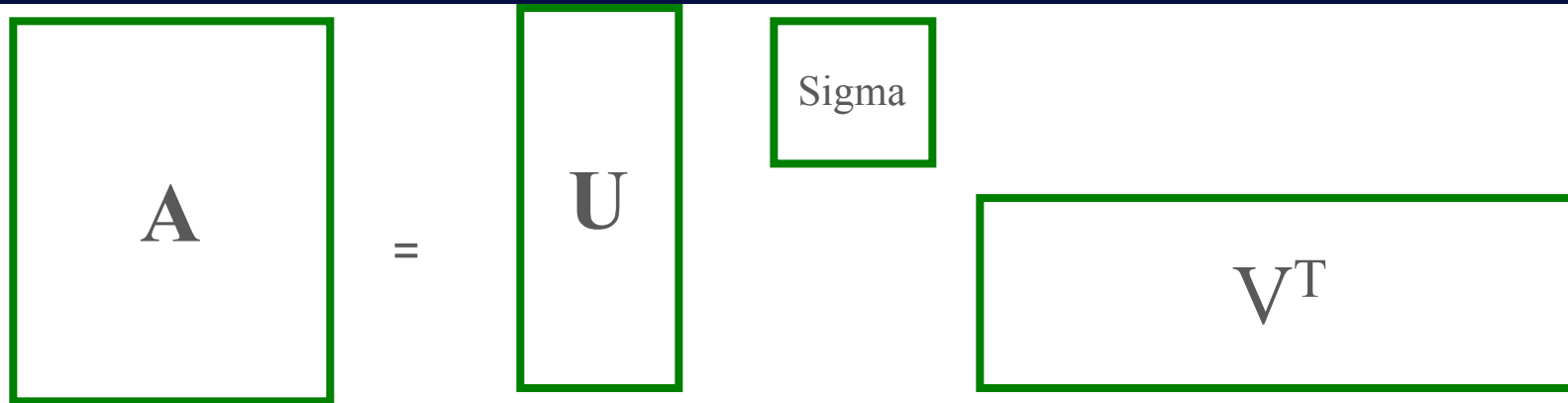
$$\|M\|_F = \sqrt{\sum_{ij} M_{ij}^2}$$

$$\|A-B\|_F = \sqrt{\sum_{ij} (A_{ij}-B_{ij})^2}$$

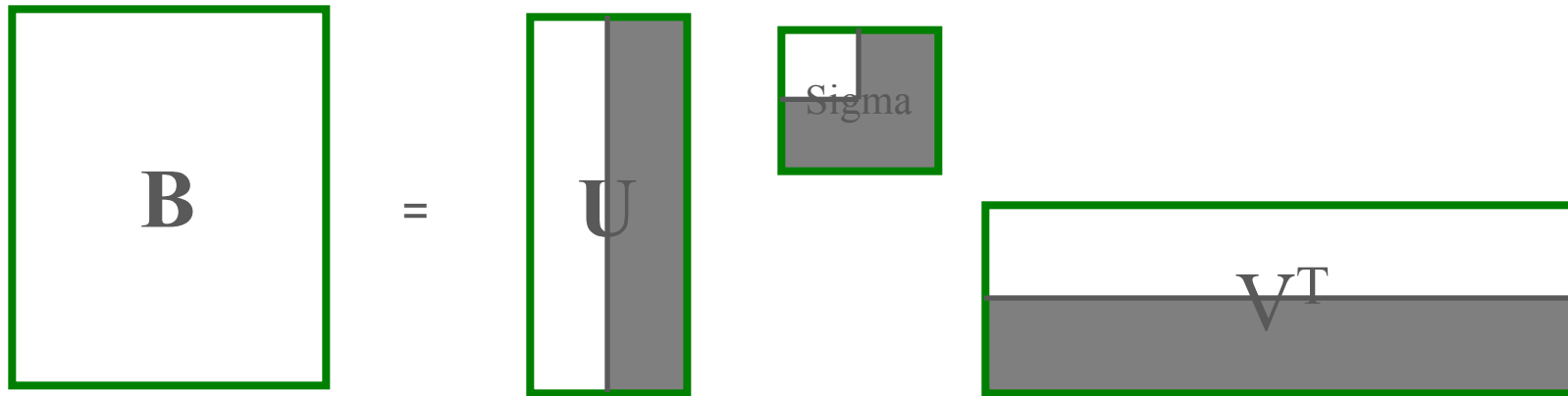
is “small”

SVD – Best Low Rank Approx.

47



B is best approximation of A



SVD – Best Low Rank Approx.

48

- Theorem:**

Let $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$ and $\mathbf{B} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ where

$\mathbf{S} = \text{diagonal } r \times r \text{ matrix}$ with $s_i = \sigma_i$ ($i=1 \dots k$) else $s_i=0$

then $\mathbf{B}$ is a **best rank($\mathbf{B}$)= k approx. to $\mathbf{A}$**

What do we mean by “best”:

- $\mathbf{B}$ is a solution to $\min_{\mathbf{B}} \|\mathbf{A} - \mathbf{B}\|_F$ where $\text{rank}(\mathbf{B})=k$

$$\begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & \\ \vdots & \vdots & \ddots & \\ x_{m1} & & & x_{mn} \end{pmatrix}_{m \times n} = \begin{pmatrix} u_{11} & \dots & u_{1r} \\ \vdots & \ddots & \vdots \\ u_{m1} & & u_{mr} \end{pmatrix}_{m \times r} \begin{pmatrix} \sigma_{11} & 0 & \dots \\ 0 & \ddots & \\ \vdots & & \end{pmatrix}_{r \times r} \begin{pmatrix} v_{11} & \dots & v_{1n} \\ \vdots & \ddots & \\ & & \end{pmatrix}_{r \times n}$$

$$\|\mathbf{A} - \mathbf{B}\|_F = \sqrt{\sum_{ij} (A_{ij} - B_{ij})^2}$$

SVD – Best Low Rank Approx.

- **Theorem:** Let $A = U \Sigma V^T$ ($\sigma_1 \geq \sigma_2 \geq \dots$, $\text{rank}(A)=r$)

then $B = U S V^T$

- S = diagonal $r \times r$ matrix where $s_i = \sigma_i$ ($i=1 \dots r$) else $s_i=0$

is a best rank- k approximation to A :

- B is a solution to $\min_B \|A-B\|_F$ where $\text{rank}(B)=k$

$$\begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & \\ \vdots & \vdots & \ddots & \\ x_{m1} & & & x_{mn} \end{pmatrix}_{m \times n} = \begin{pmatrix} u_{11} & \dots & u_{1r} \\ \vdots & \ddots & \vdots \\ u_{m1} & & u_{mr} \end{pmatrix}_{m \times r} \begin{pmatrix} \sigma_{11} & 0 & \dots \\ 0 & \ddots & \\ \vdots & & \end{pmatrix}_{r \times r} \begin{pmatrix} v_{11} & \dots & v_{1n} \\ \vdots & \ddots & \\ \vdots & & \end{pmatrix}_{r \times n}$$

- **We will need 2 facts:**

- $\|M\|_F^2 = \sum_i (q_{ii})^2$ where $M = P Q R$ is SVD of M
- $U \Sigma V^T - U S V^T = U (\Sigma - S) V^T$

SVD – Best Low Rank Approx.

- $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$, $\mathbf{B} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ ($\sigma_1 \geq \sigma_2 \geq \dots \geq 0$, $\text{rank}(\mathbf{A})=r$)
 - $\mathbf{S}$ = diagonal $n \times n$ matrix where $s_i = \sigma_i$ ($i=1 \dots k$) else $s_i=0$

then $\mathbf{B}$ is solution to $\min_{\mathbf{B}} \|\mathbf{A} - \mathbf{B}\|_F$, $\text{rank}(\mathbf{B})=k$

■ Why?

$$\min_{\mathbf{B}, \text{rank}(\mathbf{B})=k} \|\mathbf{A} - \mathbf{B}\|_F = \min \|\mathbf{\Sigma} - \mathbf{S}\|_F = \min_{s_i} \sum_{i=1}^r (\sigma_i - s_i)^2$$

$$\mathbf{U} \mathbf{\Sigma} \mathbf{V}^T - \mathbf{U} \mathbf{S} \mathbf{V}^T = \mathbf{U} (\mathbf{\Sigma} - \mathbf{S}) \mathbf{V}^T$$

- We want to choose s_i to minimize $\sum_i (\sigma_i - s_i)^2$
- Solution is to set $s_i = \sigma_i$ ($i=1 \dots k$) and other $s_i=0$

$$= \min_{s_i} \sum_{i=1}^k (\sigma_i - s_i)^2 + \sum_{i=k+1}^r \sigma_i^2 = \sum_{i=k+1}^r \sigma_i^2$$

SVD - Interpretation #2

51

Equivalent:

‘spectral decomposition’ of the matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} | & | \\ u_1 & u_2 \\ | & | \end{bmatrix} \times \begin{bmatrix} \sigma_1 & \text{ } \\ \text{ } & \sigma_2 \end{bmatrix} \times \begin{bmatrix} \text{ } & v_1 \\ \text{ } & v_2 \end{bmatrix}$$

SVD - Interpretation #2

52

Equivalent: ‘spectral decomposition’ of the matrix

$$\begin{array}{c} \begin{array}{c} \leftarrow m \rightarrow \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \\ \begin{array}{c} \uparrow \\ n \\ \downarrow \end{array} \end{array} \end{array} = \begin{array}{c} \begin{array}{c} \leftarrow k \text{ terms} \rightarrow \\ \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots \end{array} \end{array}$$

$\mathbf{u}_1$ $n \times 1$ $\mathbf{v}_1^T$ $1 \times m$

Assume: $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots \geq 0$

Why is setting small σ_i to 0 the right thing to do?

Vectors $\mathbf{u}_i$ and $\mathbf{v}_i$ are unit length, so σ_i scales them.

So, zeroing small σ_i introduces less error.

SVD - Interpretation #2

53

Q: How many σ s to keep?

A: Rule-of-a thumb:

keep 80-90% of 'energy' = $\sum_i \sigma_i^2$

$$\begin{array}{c} \updownarrow \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \downarrow \\ \text{ } \end{array} \begin{array}{c} \leftarrow \text{ } m \text{ } \rightarrow \\ \left[\begin{array}{ccccc} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{array} \right] \end{array} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots$$

Assume: $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots$

SVD - Complexity

54

- **To compute SVD:**
 - $O(nm^2)$ or $O(n^2m)$ (whichever is less)
- **But:**
 - Less work, if we just want singular values
 - or if we want first k singular vectors
 - or if the matrix is sparse

SVD - Conclusions so far

55

- **SVD:** $A = U \Sigma V^T$: **unique**
 - U : user-to-concept similarities
 - V : movie-to-concept similarities
 - Σ : strength of each concept
- **Dimensionality reduction:**
 - keep the few largest singular values (80-90% of 'energy')

Relation to Eigen-decomposition

56

- **SVD gives us:**

- $A = U \Sigma V^T$

- **Eigen-decomposition:**

- $A = X \Lambda X^T$

- A is symmetric

- U, V, X are orthonormal ($U^T U = I$),

- Λ, Σ are diagonal

- **Now let's calculate:**

- $AA^T =$

- $A^T A = V \Sigma^T U^T (U \Sigma V^T) = V \Sigma \Sigma^T V^T$

Relation to Eigen-decomposition

57

- **SVD gives us:**

- $A = U \Sigma V^T$

- **Eigen-decomposition:**

- $A = X \Lambda X^T$

- A is symmetric

- U, V, X are orthonormal ($U^T U = I$),

- Λ, Σ are diagonal

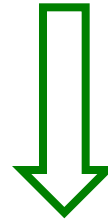
- **Now let's calculate:**

- $AA^T = U \Sigma V^T (U \Sigma V^T)^T = U \Sigma V^T (V \Sigma^T U^T) = U \Sigma \Sigma^T U^T$

- $A^T A = V \Sigma^T U^T (U \Sigma V^T) = V \Sigma \Sigma^T V^T$

$$\begin{matrix} \uparrow & \uparrow & \uparrow \\ X & \Lambda^2 & X^T \end{matrix}$$

Shows how to compute
SVD using eigenvalue
decomposition!



$$X \Lambda^2 X^T$$



Example of SVD & Conclusion



Case study: How to query?

59

- **Q: Find users that like 'Matrix'**
- **A: Map query into a 'concept space' – how?**

$$\begin{array}{c}
 \uparrow \\
 \text{SciFi} \\
 \downarrow \\
 \uparrow \\
 \text{Romnce} \\
 \downarrow
 \end{array}
 \begin{array}{c}
 \text{Matrix} \\
 \text{Alien} \\
 \text{Serenity} \\
 \text{Casablanca} \\
 \text{Amelie}
 \end{array}
 \begin{bmatrix}
 1 & 1 & 1 & 0 & 0 \\
 3 & 3 & 3 & 0 & 0 \\
 4 & 4 & 4 & 0 & 0 \\
 5 & 5 & 5 & 0 & 0 \\
 0 & 2 & 0 & 4 & 4 \\
 0 & 0 & 0 & 5 & 5 \\
 0 & 1 & 0 & 2 & 2
 \end{bmatrix}
 =
 \begin{bmatrix}
 0.13 & 0.02 & -0.01 \\
 0.41 & 0.07 & -0.03 \\
 0.55 & 0.09 & -0.04 \\
 0.68 & 0.11 & -0.05 \\
 0.15 & -0.59 & 0.65 \\
 0.07 & -0.73 & -0.67 \\
 0.07 & -0.29 & 0.32
 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix}
 12.4 & 0 & 0 \\
 0 & 9.5 & 0 \\
 0 & 0 & 1.3
 \end{bmatrix}
 \mathbf{X}
 \begin{bmatrix}
 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
 0.40 & -0.80 & 0.40 & 0.09 & 0.09
 \end{bmatrix}$$

Case study: How to query?

60

- **Q: Find users that like 'Matrix'**
- **A: Map query into a 'concept space' – how?**

$$\mathbf{q} = \begin{matrix} & \begin{matrix} \text{Matrix} \\ \text{Alien} \\ \text{Serenity} \\ \text{Casablanca} \\ \text{Amelie} \end{matrix} \\ \begin{bmatrix} 5 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Project into concept space:

Inner product with each
'concept' vector $\mathbf{v}_i$

Case study: How to query?

61

Compactly, we have:

$$\mathbf{q}_{\text{concept}} = \mathbf{q} \mathbf{V}$$

E.g.:

$$\mathbf{q} = \begin{matrix} & \begin{matrix} \text{Matrix} \\ \text{Alien} \\ \text{Serenity} \\ \text{Casablanca} \\ \text{Amelie} \end{matrix} \\ \begin{matrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \end{matrix} \end{matrix} \mathbf{x} \begin{bmatrix} 0.56 & 0.12 \\ 0.59 & -0.02 \\ 0.56 & 0.12 \\ 0.09 & -0.69 \\ 0.09 & -0.69 \end{bmatrix} = \begin{bmatrix} 2.8 & 0.6 \end{bmatrix}$$

movie-to-concept
similarities (V)

SciFi-concept
↓

Case study: How to query?

62

- How would the user d that rated ('Alien', 'Serenity') be handled?

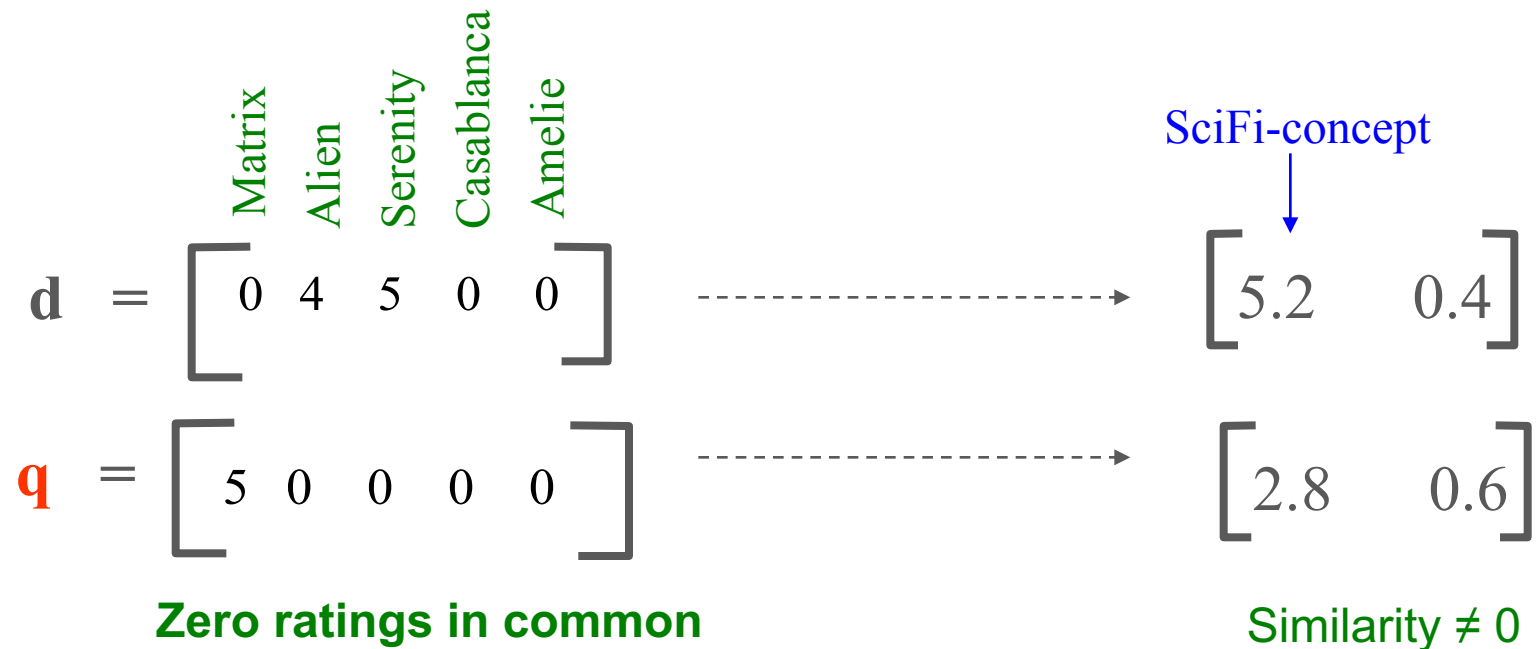
$$\mathbf{d}_{\text{concept}} = \mathbf{d} \mathbf{V}$$

E.g.:

$$\mathbf{q} = \begin{matrix} & \begin{matrix} \text{Matrix} \\ \text{Alien} \\ \text{Serenity} \\ \text{Casablanca} \\ \text{Amelie} \end{matrix} \\ \begin{bmatrix} 0 & 4 & 5 & 0 & 0 \end{bmatrix} & \mathbf{X} \end{matrix} \begin{matrix} \begin{bmatrix} 0.56 & 0.12 \\ 0.59 & -0.02 \\ 0.56 & 0.12 \\ 0.09 & -0.69 \\ 0.09 & -0.69 \end{bmatrix} \\ \text{movie-to-concept} \\ \text{similarities (V)} \end{matrix} = \begin{matrix} \text{SciFi-concept} \\ \downarrow \\ \begin{bmatrix} 5.2 & 0.4 \end{bmatrix} \end{matrix}$$

Case study: How to query?

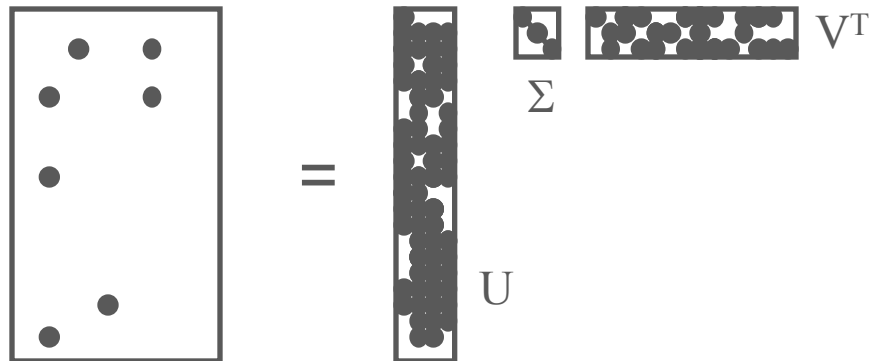
- **Observation:** User d that rated (*Alien*, *Serenity*) will be **similar** to user q that rated (*Matrix*), although d and q have **zero ratings in common**!



SVD: Drawbacks

64

- + **Optimal low-rank approximation**
in terms of Frobenius norm
- **Interpretability problem:**
 - A singular vector specifies a linear combination of all input columns or rows
- **Lack of sparsity:**
 - Singular vectors are **dense!**



CUR Decomposition



Frobenius norm:

$$\|X\|_F = \sqrt{\sum_{ij} X_{ij}^2}$$

CUR Decomposition

66

- **Goal: Express A as a product of matrices C,U,R**

Make $\|A - C \cdot U \cdot R\|_F$ small

- **“Constraints” on C and R:**

$$\left(\begin{array}{c} \text{Red} \\ \text{Yellow} \\ \text{Grey} \end{array} \right) \approx \left(\begin{array}{c} \text{Red} \\ \text{Red} \\ \text{Red} \\ \text{Yellow} \\ \text{Grey} \\ \text{Grey} \end{array} \right) \cdot \left(\begin{array}{c} U \\ \end{array} \right) \cdot \left(\begin{array}{c} R \\ \end{array} \right)$$

A C U R

$$\|X\|_F = \sqrt{\sum_{ij} X_{ij}^2}$$

CUR Decomposition

67

- **Goal: Express A as a product of matrices C,U,R**

Make $\|A - C \cdot U \cdot R\|_F$ small

- **“Constraints” on C and R:**

$$\begin{pmatrix} \text{red bar} \\ \text{gray bar } A \\ \text{yellow bar} \end{pmatrix} \approx \begin{pmatrix} C \end{pmatrix} \cdot \begin{pmatrix} U \\ \text{red arrow} \end{pmatrix} \cdot \begin{pmatrix} \text{red bar} \\ \text{red bar} \\ \text{red bar} \\ \text{gray bar } R \\ \text{gray bar} \\ \text{yellow bar} \end{pmatrix}$$

A C U R

Pseudo-inverse of
the intersection of C and R

CUR: How it Works

68

- **Sampling columns (similarly for rows):**

Input: matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, sample size c

Output: $\mathbf{C}_d \in \mathbb{R}^{m \times c}$

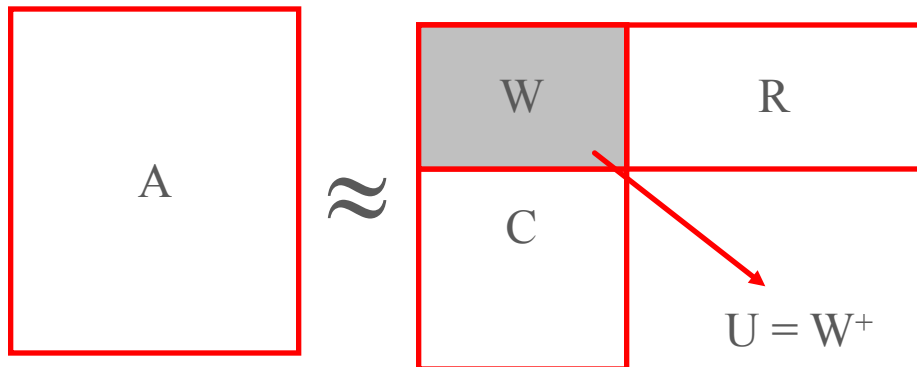
1. for $x = 1 : n$ [column distribution]
2. $P(x) = \sum_i \mathbf{A}(i, x)^2 / \sum_{i,j} \mathbf{A}(i, j)^2$
3. for $i = 1 : c$ [sample columns]
4. Pick $j \in 1 : n$ based on distribution $P(x)$
5. Compute $\mathbf{C}_d(:, i) = \mathbf{A}(:, j) / \sqrt{cP(j)}$

Note this is a randomized algorithm, same column can be sampled more than once

Computing U

69

- Let $\mathbf{W}$ be the “intersection” of sampled columns $\mathbf{C}$ and rows $\mathbf{R}$
 - Let SVD of $\mathbf{W} = \mathbf{X} \mathbf{Z} \mathbf{Y}^T$
- **Then:** $\mathbf{U} = \mathbf{W}^+ = \mathbf{Y} \mathbf{Z}^+ \mathbf{X}^T$
 - $\mathbf{Z}^+$: **reciprocals of non-zero singular values:** $Z_{ii}^+ = 1/Z_{ii}$
 - $\mathbf{W}^+$ is the “**pseudoinverse**”



Why pseudoinverse works?

$\mathbf{W} = \mathbf{X} \mathbf{Z} \mathbf{Y}$ then $\mathbf{W}^{-1} = \mathbf{X}^{-1} \mathbf{Z}^{-1} \mathbf{Y}^{-1}$

Due to orthonormality

$\mathbf{X}^{-1} = \mathbf{X}^T$ and $\mathbf{Y}^{-1} = \mathbf{Y}^T$

Since $\mathbf{Z}$ is diagonal $\mathbf{Z}^{-1} = 1/Z_{ii}$

Thus, if $\mathbf{W}$ is nonsingular,
pseudoinverse is the true
inverse

CUR: Provably good approx. to SVD

70

- **For example:**

- Select $c = O\left(\frac{k \log k}{\epsilon^2}\right)$ columns of A using **ColumnSelect** algorithm
- Select $r = O\left(\frac{k \log k}{\epsilon^2}\right)$ rows of A using **ColumnSelect** algorithm
- Set $U = W^+$

- **Then:**

with probability 98%

$$\|A - \overset{\text{CUR error}}{CUR}\|_F \leq (2 + \epsilon) \|A - \overset{\text{SVD error}}{A_k}\|_F$$

In practice:

Pick $4k$ cols/rows

for a “rank- k ” approximation

CUR: Pros & Cons

71

+ Easy interpretation

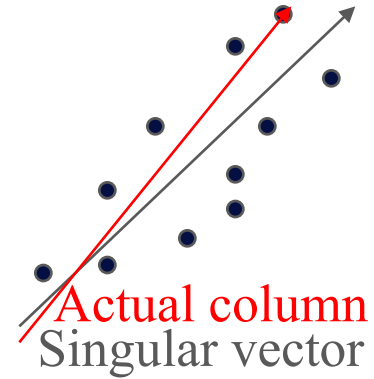
- Since the basis vectors are actual columns and rows

+ Sparse basis

- Since the basis vectors are actual columns and rows

- Duplicate columns and rows

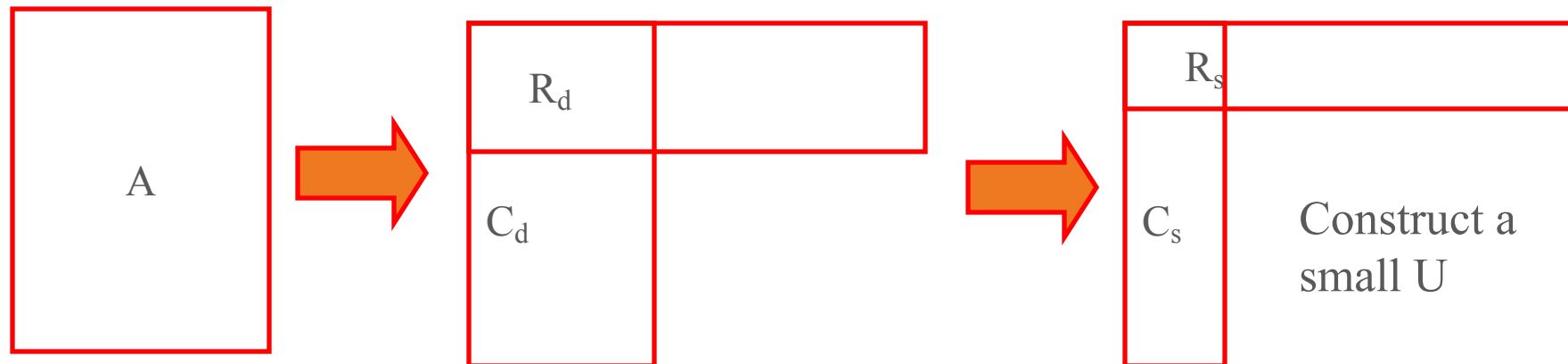
- Columns of large norms will be sampled many times



Solution

72

- If we want to get rid of the duplicates:
 - Throw them away
 - Scale (multiply) the columns/rows by the square root of the number of duplicates



SVD vs. CUR

73

SVD: $A = U \Sigma V^T$

Annotations for SVD:

- A : Huge but sparse
- U : Big and dense
- Σ : sparse and small
- V^T : Big and dense

CUR: $A = C U R$

Annotations for CUR:

- A : Huge but sparse
- C : Big but sparse
- U : dense but small
- R : Big but sparse

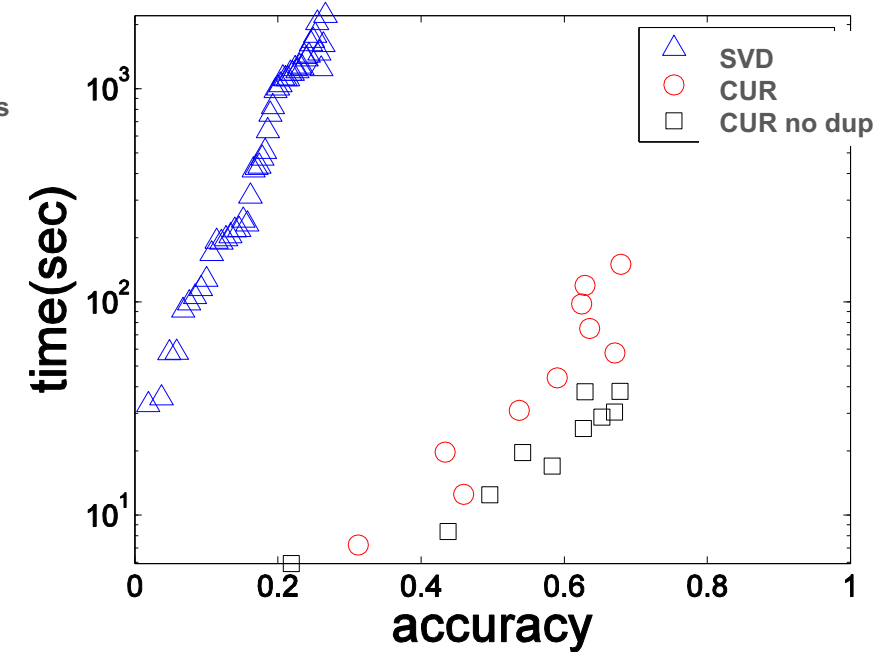
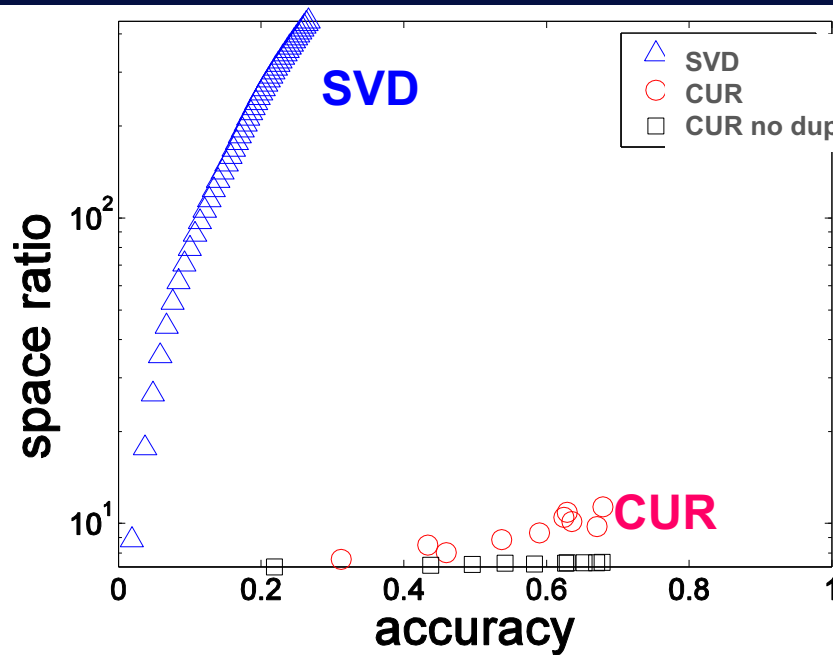
SVD vs. CUR: Simple Experiment

74

- **DBLP bibliographic data**
 - Author-to-conference big sparse matrix
 - A_{ij} : Number of papers published by author i at conference j
 - 428K authors (rows), 3659 conferences (columns)
 - **Very sparse**
- **Want to reduce dimensionality**
 - How much time does it take?
 - What is the reconstruction error?
 - How much space do we need?

Results: DBLP- big sparse matrix

75



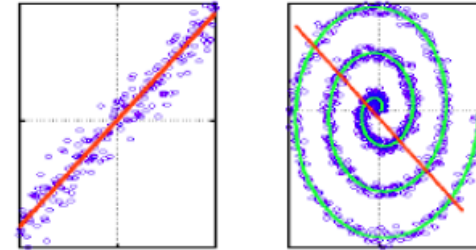
- **Accuracy.**
 - $1 - \text{relative sum squared errors}$
- **Space ratio:**
 - $\text{\#output matrix entries} / \text{\#input matrix entries}$
- **CPU time**

What about linearity assumption?

76

- **SVD is limited to linear projections:**

- Lower-dimensional linear projection that preserves Euclidean distances

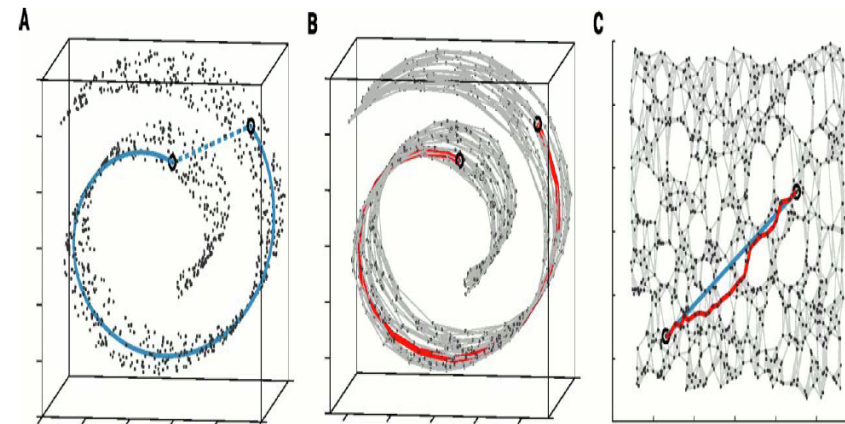


- **Non-linear methods: Isomap**

- Data lies on a nonlinear low-dim curve aka manifold
 - Use the distance as measured along the manifold

- **How?**

- Build adjacency graph
- Geodesic distance is graph distance
- SVD/PCA the graph pairwise distance matrix



Basic operations on matrices



Types of matrices

Row matrix or vector

Any number of columns but only one row

$$\begin{bmatrix} 1 & 1 & 6 \end{bmatrix} \quad \begin{bmatrix} 0 & 3 & 5 & 2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \end{bmatrix}$$

Column matrix or vector

Number of rows may be any integer but the number of columns is always 1

$$\begin{matrix} 1 & a_{11} \\ 4 & a_{21} \\ & \vdots \\ 2 & a_{m1} \end{matrix}$$

Types of matrices

Diagonal matrix

A square matrix where all the elements are zero except those on the main diagonal

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Square matrix

Number of rows = number of columns

$$\begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 9 \end{pmatrix}$$

Types of matrices

Unit or Identity matrix - I

A diagonal matrix with ones on the main diagonal

$$\begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{matrix} \quad \begin{matrix} a_{ij} & 0 \\ 0 & a_{ij} \end{matrix}$$

Null (zero) matrix - 0

All elements in the matrix are zero

$$\begin{matrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{matrix}$$

Types of matrices

Triangular matrix

A square matrix whose elements above or below the main diagonal are all zero

1	0	0
2	1	0
5	2	3

1	8	9
0	1	6
0	0	3

Upper/Lower triangular matrix

A square matrix whose elements below/above the main diagonal are all zero

Types of matrices

Scalar matrix

A diagonal matrix whose main diagonal elements are equal to the same scalar

A scalar is defined as a single number or constant

A square matrix whose elements above or below the main diagonal are all zero

$$\begin{matrix} a_{ij} & 0 & 0 \\ 0 & a_{ij} & 0 \\ 0 & 0 & a_{ij} \end{matrix}$$

$$\begin{matrix} 1 & 0 & 0 & 6 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 6 \end{matrix}$$

Addition and Subtraction of matrices

- Must be of the same order, mxn:

$$A + B = a_{ij} + b_{ij}$$

$$\begin{bmatrix} 4 & 3 & 1 \\ 0 & 5 & 2 \\ 5 & 6 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 6 & 7 & 9 \\ 0 & 4 & 8 \end{bmatrix}$$

$$A - B = a_{ij} - b_{ij}$$

Addition

$$\begin{bmatrix} 5 & 5 & 4 \\ 6 & 12 & 11 \\ 5 & 10 & 8 \end{bmatrix}$$

Substraction

$$\begin{bmatrix} 3 & 1 & -2 \\ -6 & -2 & -7 \\ 5 & 2 & -8 \end{bmatrix}$$

Scalar Multiplication

- The **scalar product** of a number k and a matrix A
- k : **scalar**.

$$kA = ka_{ij}$$

For all i and j $a_{ij} = a_{ji}^T$

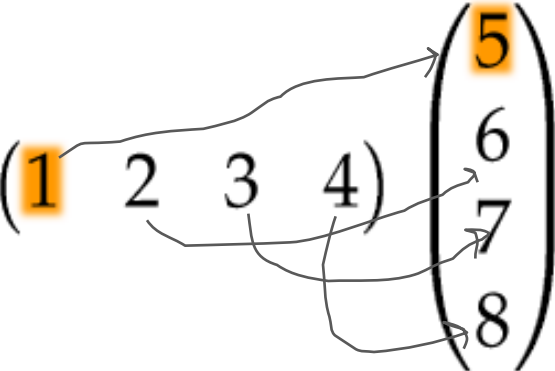
Transpose of a matrix

$$A = \begin{pmatrix} 2 & 4 & 7 \\ 5 & 3 & 1 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 2 & 5 \\ 4 & 3 \\ 7 & 1 \end{pmatrix}$$

Product of a Row Matrix and a Column Matrix

- Number of columns of a row matrix = the number of rows of a column matrix
- $R(1 \times 4) \times C(4 \times 1) = M(1 \times 1)$


$$(1 \ 2 \ 3 \ 4) \begin{pmatrix} 5 \\ 6 \\ 7 \\ 8 \end{pmatrix} = 1 \cdot 5 + 2 \cdot 6 + 3 \cdot 7 + 4 \cdot 8 = 70$$

Revenue of a car dealer

- A car dealer sells four model types: A,B,C,D. On a given week, this dealer sold 10 cars of model A, 5 of model B, 8 of model C and 3 of model D. The selling prices of each automobile are respectively \$12,500, \$11,800, \$15,900 and \$25,300. Represent the data using matrices and use matrix multiplication to find the total revenue.

$$\begin{bmatrix} 10 & 5 & 8 & 3 \end{bmatrix} \begin{bmatrix} 12,500 \\ 11,800 \\ 15,900 \\ 25,300 \end{bmatrix} = [10(12,500) + 5(11,800) + 8(15,900) + 3(25,300)] = [387,100]$$

Matrix Product

- **A** is an $m \times p$ matrix
- **B** is a $p \times n$ matrix,
- **Matrix product** of A and B denoted by AB is an $m \times n$ matrix

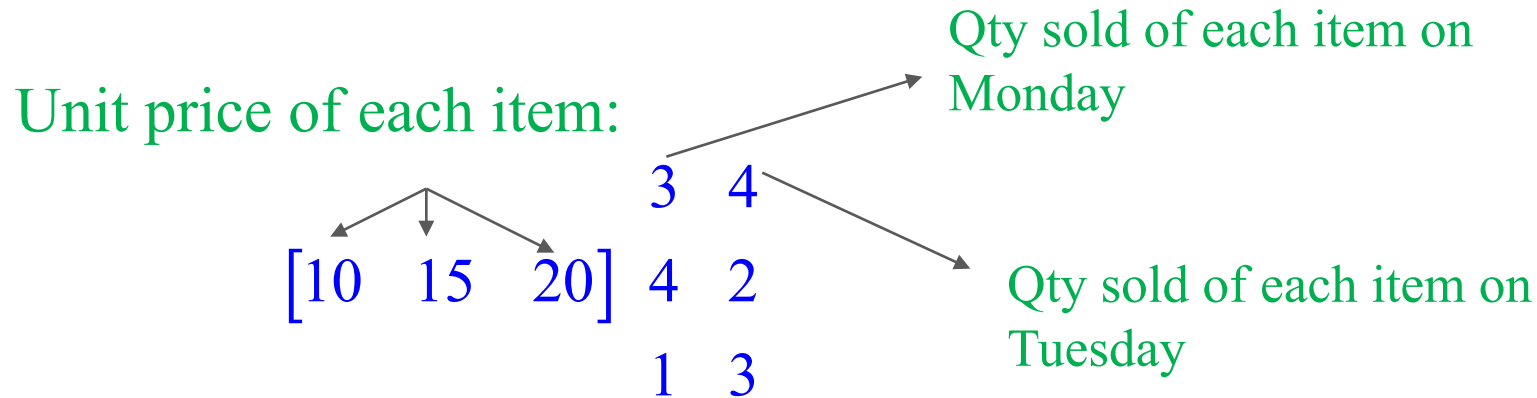
Is Matrix Multiplication Commutative?
 $A(3 \times 4)B(4 \times 3) = BA$?

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix} = \begin{pmatrix} 70 & 80 & 90 \\ 158 & 184 & 210 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{pmatrix} \text{ is not defined}$$

Practical application

- Suppose you a business owner and sell clothing. The following represents the number of items sold and the cost for each item: Use matrix operations to determine the total revenue over the two days:
- Monday: 3 T-shirts at \$10 each, 4 hats at \$15 each, and 1 pair of shorts at \$20.
Tuesday: 4 T-shirts at \$10 each, 2 hats at \$15 each, and 3 pairs of shorts at \$20.



Orthogonal matrices (orthonormal)

- Each column vector is orthogonal to every other column vector in the matrix.
- Each column in the matrix has length 1
- $M^T M = I$
- Columns of M form an orthonormal basis of $\mathbb{R}^n$

$$\begin{bmatrix} -0.8 & 0.6 \\ 0.6 & 0.8 \end{bmatrix}$$

Product AB of two orthogonal $n \times n$ matrices is orthogonal

1	3
3	5
5	3
0	4
1	5



THANK YOU !

Center Of Computer Engineering

www.cce.hcmut.edu.vn